

BRAC University
Department of Electrical and Electronic Engineering
EEE/ECE-103IL: Computer Programming Laboratory
Spring 2026 (Section: 02)

Lab Final

1. Develop a **C program** that dynamically allocates memory to store the weights of **N packages** to be loaded onto a cargo truck. The value of **N** and the package weights will be entered by the user. [CO4] (10)

The package weights are recorded in the order they are received.

To maintain proper balance of the truck:

- **Heavier packages must be loaded first**
- **Lighter packages must be loaded later**

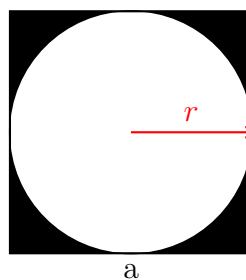
Write a C program that performs the following tasks:

- Dynamically allocate memory to store the package weights
- Read all input values from the user
- Process the data according to the loading rule
- Display the final loading sequence

Constraints:

- Dynamic memory allocation mandatory
- Do not use any built-in functions

2. The diagram below shows a circle inscribed in a square with both shapes sharing the same center. [CO4] (8)



Assume a is the side length of the square, provided by the user. Write a C program using a function to:

- Calculate the area of the square and circle
- Calculate the percentage of the square's area covered by the inscribed circle
- Print the result upto two decimal places

Use the formula:

$$\text{Area of circle} = \pi r^2$$

3. The following program is supposed to print **all integers** that are not divisible by 3 and 5 and lie between 1 and a user-defined number (inclusive). The program also counts the total **number** of such integers and prints the result. [CO4] (7)

No.	Code	Explanation/Corrected
1		
2	int main() {	
3	int end;	
4	printf("Enter the end number: ");	
5	scanf("%f", &end);	
6	int i, Count;	
7	count = 0;	
8	for(i=1; i<end; i++) {	
9	if ((i/3 != 0) & (i/5 != 0)) {	
10	count = count + 1;	
11	printf("%d\n", i)	
12	}	
13		
14	printf("Total count is: %c\n", Count);	
15	}	